PHYS 225

Week 3, Lecture 3: Vectors, Matrices, and Index Notation

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

Reminders

- HW 2 due (HW 1 late due) at 11:59 pm this Thursday, 2/5
- Office Hours **tomorrow 2/4 4-7 pm, Thursday 2/5 5-7 pm.** To receive bonus points for OHs attendance – sign in, take a photo of your signature, upload to Gradescope.
- Don’t forget to turn in your **1-minute papers** by 6pm today!

Week 3, Lecture 3
Vectors, Matrices, and Index Notation

Objectives:

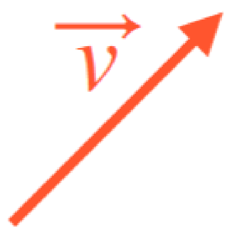
- Review some vector manipulation
- Review how to multiply vectors by matrices
- Explain why linear transformations are used in physics
- Learn index notation and the Einstein summation convention

One Minute Paper: Give two examples of physical quantities that can be represented by vectors

Week 3, Lecture 3
Vectors

Simple notion of a vector: object with magnitude and direction

Some common vector quantities in physics: force, acceleration, velocity, displacement, angular momentum



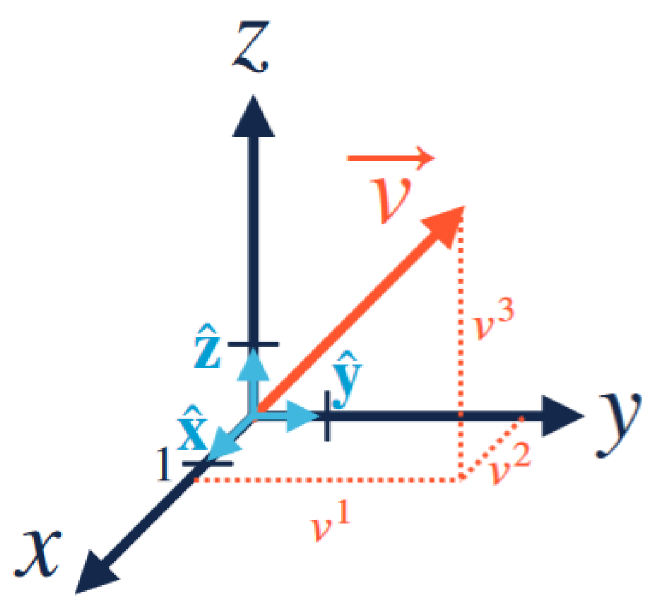
Week 3, Lecture 3
Unit vectors and components

In a coordinate system, can decompose in terms of unit vectors:

$$\vec{v} = v^1\hat{\mathbf{x}} + v^2\hat{\mathbf{y}} + v^3\hat{\mathbf{z}}$$

This lets us define a vector by its components:

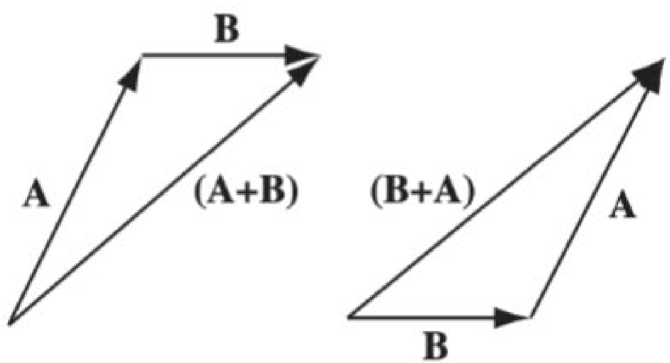
$$\vec{v} = (v^1, v^2, v^3)$$



Week 3, Lecture 3
Addition and scalar multiplication

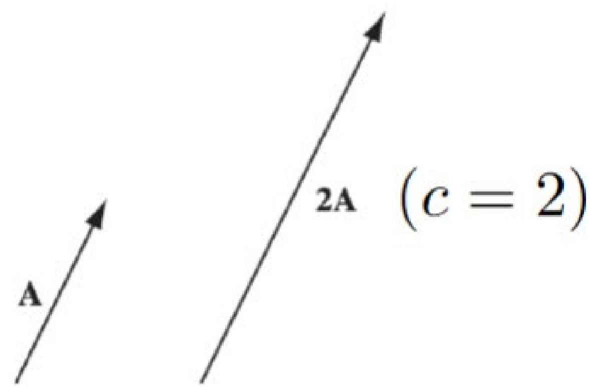
Vectors can be added:

$$\vec{A} + \vec{B} = (A^1 + B^1)\hat{\mathbf{x}} + (A^2 + B^2)\hat{\mathbf{y}} + (A^3 + B^3)\hat{\mathbf{z}}$$



Vectors can be multiplied by a scalar (real number):

$$\vec{A}' = c\vec{A}$$



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Dot product

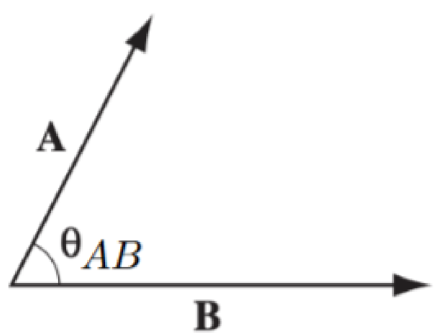
Two vectors can also be “multiplied” together using the **dot product**

Dot product takes two vectors and outputs a scalar:

$$\begin{aligned}\overrightarrow{A} \cdot \overrightarrow{B} &= A^1B^1 + A^2B^2 + A^3B^3 \\ &= |A|\,|B|\cos\theta_{AB}\end{aligned}$$

magnitude of A is the length of the vector:

$$|A| = \sqrt{(A^1)^2 + (A^2)^2 + (A^3)^2}$$



Week 3, Lecture 3
Matrices

Matrices allow us to transform vectors.

An $m \times n$ matrix is a rectangular array of numbers with m rows and n columns.

matrix entries: $M^{i}{}_{j}$ labels row labels column

$$M = \begin{pmatrix} M^1{}_1 & M^1{}_2 & \dots & M^1{}_n \\ M^2{}_1 & M^2{}_2 & \dots & M^2{}_n \\ \vdots & & \ddots & \\ M^m{}_1 & & & M^m{}_n \end{pmatrix}$$

$i = 1, \dots, m$ and $j = 1, \dots, n$

In PHYS 225 we'll only see square matrices! $n = m$

Week 3, Lecture 3
Multiplying a vector by a matrix

We can transform a vector by multiplying it by a matrix: $\overrightarrow{b} = \mathsf{A}\overrightarrow{x}$

$$\mathsf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \qquad \overrightarrow{x} = \begin{pmatrix} x^1 \\ x^2 \end{pmatrix}$$

note: we often write vectors as rows to save space, but to multiply by a matrix, use a column vector

$$\overrightarrow{b} = \mathsf{A}\overrightarrow{x} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x^1 \\ x^2 \end{pmatrix}$$

Week 3, Lecture 3
Multiplying a vector by a matrix

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$$\overrightarrow{b} = \mathrm{A}\overrightarrow{x} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x^1 \\ x^2 \end{pmatrix} = \begin{pmatrix} ax^1 + bx^2 \\ \cdots \end{pmatrix}$$

Week 3, Lecture 3
Multiplying a vector by a matrix

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Week 3, Lecture 3
Matrices as linear transformations

We can also write our matrix multiplication as a system of linear equations:

$$\begin{pmatrix} b^1 \\ b^2 \end{pmatrix} = \vec{b} = \mathsf{A}\vec{x} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x^1 \\ x^2 \end{pmatrix} = \begin{pmatrix} ax^1 + bx^2 \\ cx^1 + dx^2 \end{pmatrix}$$

$$\begin{aligned} b^1 &= ax^1 + bx^2 \\ b^2 &= cx^1 + dx^2 \end{aligned}$$

Multiplying a vector by a matrix gives a **linear combination** of the vector's components

Week 3, Lecture 3
Identity matrix

There are special matrices that we will often use:

The identity matrix: $$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Note: The subscript indicates the size of the matrix. Often people just write I

The identity matrix takes any vector and returns the same vector:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x^1 \\ x^2 \\ x^3 \end{pmatrix} = \begin{pmatrix} x^1 \\ x^2 \\ x^3 \end{pmatrix}$$

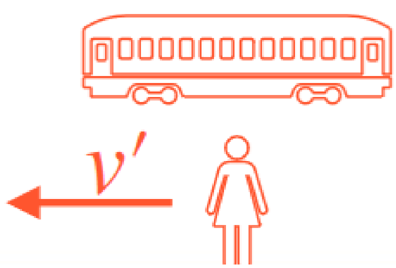
Week 3, Lecture 3
Physics and linear algebra

In Newtonian physics, the Galilean transformations (Discussion 1) are an example of **linear** coordinate transformations:

$$\Delta t = \Delta t'$$
$$\Delta x = v\Delta t' + \Delta x'$$

This only works because our frames agree on what (inertial) non-accelerating motion is:

$$\Delta x = v\Delta t \qquad\qquad \Delta x' = v'\Delta t'$$



Physics must be invariant under different inertial coordinate systems
$\Rightarrow$ Coordinate transformations must be linear in x and t
(Morin pp. 72-23)

Week 3, Lecture 3
Index notation and Kronecker delta

Instead of writing out the individual components of a vector or matrix, there is a compact notation that lets us write them all at the same time

Let's define the Kronecker delta:

$$\delta^i_j = \begin{cases} 0 \text{ if } i \neq j \\ 1 \text{ if } i = j \end{cases}$$

What does this look like in matrix form?
If $i, j = 1,2,3$, we have a 3 x 3 matrix:

$$\delta^i_j = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Just a fancy way of writing the identity matrix!

Week 3, Lecture 3
Matrix multiplication with index notation

Let's multiply a vector by the (2 x 2) identity matrix in index notation:

$$\overrightarrow{v} = I\overrightarrow{v}$$

$$\begin{pmatrix} v^1 \\ v^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} v^1 \\ v^2 \end{pmatrix}$$

We can write this compactly as a sum:

$$v^i = \sum_{j=1}^{2} \delta^i_j v^j$$

means

$$v^1 = \delta^1_1 v^1 + \delta^1_2 v^2$$
$$v^2 = \delta^2_1 v^1 + \delta^2_2 v^2$$

Week 3, Lecture 3
Einstein summation notation

In every linear algebra multiplication, something is being summed.

The Einstein summation convention says "let's agree to not write the summation symbol to save time."

$$v^i = \sum_{j=1}^{2} \delta^i_j v^j \longrightarrow v^i = \delta^i_j v^j$$

the repeated "dummy"index is summed over

IMPORTANT! $v^i$and v^j both stand for the same vector $\vec{v}$,
but to avoid confusion, we always choose different indices if not summed over

Week 3, Lecture 3
Matrix-vector multiplication: Einstein notation

Write $\overrightarrow{v} = M\overrightarrow{x}$ in Einstein summation notation:

$$M = \begin{pmatrix} M^1_{\ 1} & M^1_{\ 2} \\ M^2_{\ 1} & M^2_{\ 2} \end{pmatrix} \quad \overrightarrow{x} = \begin{pmatrix} x^1 \\ x^2 \end{pmatrix}$$

Week 3, Lecture 3
Matrix-vector multiplication: Einstein notation

Write $\vec{v} = M\vec{x}$ in Einstein summation notation:

$$M = \begin{pmatrix} M^1{}_1 & M^1{}_2 \\ M^2{}_1 & M^2{}_2 \end{pmatrix} \qquad \vec{x} = \begin{pmatrix} x^1 \\ x^2 \end{pmatrix}$$

"free" (un-summed) indices match on each side

$$v^1 = M^1{}_1 x^1 + M^1{}_2 x^2$$

$$v^2 = M^2{}_1 x^1 + M^2{}_2 x^2$$

$$\longrightarrow \quad v^i = M^i{}_j\, x^j$$

repeated index summed over "northeast"

Week 3, Lecture 3
Index notation for operators

Matrices eat vectors and spit out vectors:

$$M^i{}_j : v^j \to v^i$$

In linear algebra, we would call this a **linear operator**

Let's define an operator that eats two vectors and spits out a **number**:

$$\delta_{ij} : v^i, v^j \to \mathbb{R}$$

(we also call this the Kronecker delta)

Note: $\delta_{ij} = \begin{cases} 0 \text{ if } i \neq j \\ 1 \text{ if } i = j \end{cases}$ has the same numerical components as δ^i_j. But it has both indices downstairs, so it eats 2 vectors

Week 3, Lecture 3
Dot product in index notation

Recall the dot product of

$\overrightarrow{A} = (A^1, A^2, A^3)$ and $\overrightarrow{B} = (B^1, B^2, B^3)$:

$$\overrightarrow{A} \cdot \overrightarrow{B} = A^1B^1 + A^2B^2 + A^3B^3 = \sum_{i=1}^{3} A^iB^i$$

We can write this even more compactly using the new Kronecker delta we just defined:

$$\overrightarrow{A} \cdot \overrightarrow{B} = \sum_{i=}^{3} \sum_{j=1}^{3} \delta_{ij} A^i B^j$$

UGH! A double sum?? But it's really not so bad...

Week 3, Lecture 3
Dot product in index notation

Write in Einstein Summation Notation

$$\vec{A} \cdot \vec{B} = \sum_{i=1}^{3} \sum_{j=1}^{3} \delta_{ij} A^i B^j$$

$$= \delta_{ij} A^i B^j$$

(two summations = two repeated indices)

Week 3, Lecture 3
Dot product in index notation

Write in Einstein Summation Notation

$$\vec{A}\cdot\vec{B} = \sum_{i=1}^{3}\sum_{j=1}^{3}\delta_{ij}A^iB^j$$

$$= \delta_{ij}A^iB^j$$ (two summations = two repeated indices)

$$= \delta_{11}A^1B^1 + \cancelto{0}{\delta_{12}}A^1B^2 + \cancelto{0}{\delta_{13}}A^1B^3$$
$$+\cancel{\delta_{21}}A^2B^1 + \delta_{22}A^2B^2 + \cancel{\delta_{23}}A^2B^3$$
$$+\cancel{\delta_{31}}A^3B^1 + \cancel{\delta_{32}}A^3B^2 + \delta_{33}A^3B^3$$

$$(\delta_{11} = \delta_{22} = \delta_{33} = 1)$$

The dot product!

$$\boxed{\vec{A}\cdot\vec{B} = A^1B^1 + A^2B^2 + A^3B^3}$$

Week 3, Lecture 3
Rules of thumb for index notation

1. Never repeat an index more than once!

$$\delta_{ij} v^i w^i = ???$$

2. To translate to (ordinary) matrix notation, all summations are "northeast":

$$M\vec{v} \equiv M^i{}_j v^j$$

3. But once you write the indices, order doesn't matter:

$$M^i{}_j v^j = v^j M^i{}_j = M\vec{v}, \text{ NOT } \vec{v}M!$$

4. Free (un-summed) indices must match on both sides:

$$\vec{a} = M\vec{b} \Leftrightarrow a^i = M^i{}_j b^j$$

Week 3, Lecture 3
Vectors, Matrices, and Index Notation

Summary

- Many physical quantities can be represented as vectors
- Linear coordinate transformations relate physics in different inertial reference frames
- Matrix manipulations allow us to encode these linear transformations of coordinate systems
- Index notation is a compact way to represent linear algebra calculations

One Minute Paper: What types of calculations would the Einstein summation convention be useful for?